BIOS 735: Linear Regression

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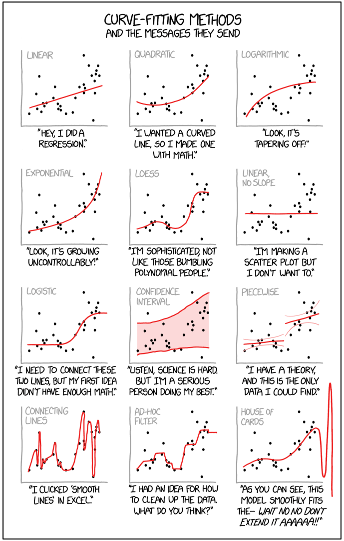
Outline


CURVE-FITTING METHODS
AND THE MESSAGES THEY SEND
LINEAR
"HEY, I DID A REGRESSION."
QUADRATIC
"I WANTED A CURVED LINE, SO I MADE ONE WITH MATH."
LOGARITHMIC
"LOOK, IT'S TAPERING OFF!"
EXPONENTIAL
"LOOK, IT'S GROWING UNCONTROLLABLY!"
LOESS
"I'M SOPHISTICATED, NOT LIKE THOSE BUMBLING POLYNOMIAL PEOPLE."
LINEAR, NO SLOPE
"I'M MAKING A SCATTER PLOT BUT I DON'T WANT TO."
LOGISTIC
"I NEED TO CONNECT THESE TWO LINES, BUT MY FIRST IDEA DIDN'T HAVE ENOUGH MATH."
CONFIDENCE INTERVAL
"LISTEN, SCIENCE IS HARD. BUT I'M A SERIOUS PERSON DOING MY BEST."
PIECEWISE
"I HAVE A THEORY, AND THIS IS THE ONLY DATA I COULD FIND."
CONNECTING LINES
"I CLICKED 'SMOOTH LINES' IN EXCEL."
AD-HOC FILTER
"I HAD AN IDEA FOR HOW TO CLEAN UP THE DATA. WHAT DO YOU THINK?"
HOUSE OF CARDS
"AS YOU CAN SEE, THIS MODEL SMOOTHLY FITS THE– WAIT NO NO DON'T EXTEND IT AAAAAA!!"

Linear Regression

- Linear regression models were developed before computers were around, but are still popular today.
- They are simple and (commonly) adequate and interpretable representations of the real world.
- They commonly work as well as difficult non-linear models for prediction, especially when there is limited data and low signal-to-noise ratio.
- Linear models do have assumptions, but many properties are robust to these assumptions (plus we can transform outcomes).

Linear Regression model

- A linear regression model is defined as

$$
\begin{aligned}
Y &= \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon \\
&= \beta_0 + \sum_{j=1}^{p} \beta_j X_j + \epsilon \\
&= f(\boldsymbol{X}) + \epsilon
\end{aligned}
$$

  - Y is our outcome of interest
  - $(X_1, X_2 \ldots, X_p)$ are our covariates of interest
  - β_0: intercept;
  - $\beta_j, j = 1, \cdots, p$: regression coefficients.
  - $E(\epsilon) = 0$ and $Var(\epsilon) = \sigma^2$.

Linear Regression model

- We can also define it in matrix form with the following

$$\boldsymbol{X} = \begin{pmatrix} 1 & X_{11} & \dots & X_{1p} \\ 1 & X_{21} & \dots & X_{2p} \\ \vdots & \vdots & & \\ 1 & X_{n1} & \dots & X_{np} \end{pmatrix} \quad \boldsymbol{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} \quad \boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix} \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}$$

  as the covariate data, observations, error term, and parameters in a linear regression model.
- The model can be expressed succinctly as

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

Least squares and RSS

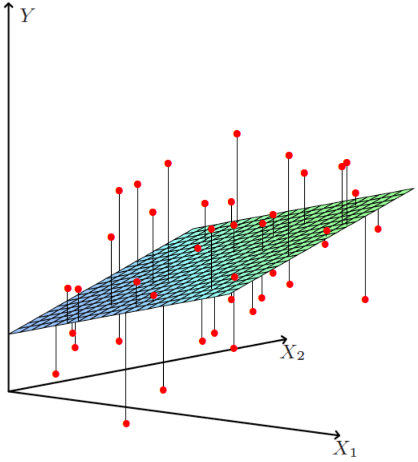
- The residual sum of squares for β is defined as

$$
\begin{aligned}
RSS(\beta) &= \sum_{i=1}^{n} (y_i - \boldsymbol{X}_i'\beta)^2 \\
&= \sum_{i=1}^{n} \{y_i - f(\boldsymbol{X}_i)\}^2 \\
&= (\boldsymbol{y} - \boldsymbol{X}\beta)'(\boldsymbol{y} - \boldsymbol{X}\beta) = ||\boldsymbol{y} - \boldsymbol{X}\beta||^2
\end{aligned}
$$

  where $\boldsymbol{X}_i = (X_{i1}, X_{i2}, \ldots, X_{ip})'$

- Using least squares criteria the estimate of β, denoted by $\hat{\beta}$, is the value that minimizes the RSS.

Least squares and RSS



The Least Squares Solution

- To minimize $RSS(\beta) = (\boldsymbol{y} - \boldsymbol{X}\beta)'(\boldsymbol{y} - \boldsymbol{X}\beta)$ we differentiate with respect to β to get

$$\begin{aligned} \frac{\partial RSS(\beta)}{\partial \beta} &= -2\boldsymbol{X}'(\boldsymbol{y} - \boldsymbol{X}\beta) \\ \frac{\partial^2 RSS(\beta)}{\partial \beta \partial \beta'} &= 2\boldsymbol{X}'\boldsymbol{X}. \end{aligned}$$

- If $\boldsymbol{X}$ has full column rank, setting the first derivative to zero yields,

$$\begin{aligned} 0 &= \boldsymbol{X}'(\boldsymbol{y} - \boldsymbol{X}\beta) \quad \text{and} \\ \hat{\beta} &= (\boldsymbol{X}'\boldsymbol{X})^{-1}X'\boldsymbol{y}. \end{aligned}$$

Predicted values

- Predicted values at a point $\boldsymbol{x}_0$ are given by $\hat{f}(\boldsymbol{x}_0) = \boldsymbol{x}_0'\hat{\beta}$.
- The predicted values for the training data are given by

$$\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\beta} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}.$$

- $H = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$ is referred to as the "Hat Matrix"
- Geometrically, $\hat{\beta}$ is chosen such that $\boldsymbol{y} - \hat{\boldsymbol{y}}$ is orthogonal to the subspace of $\boldsymbol{X}$.

Rank Deficiencies

- Recall, we said *"if $\boldsymbol{X}$ has full column rank"*
- When might this not be the case?

"Assumptions" standard regression models

Linear, Independent, Normality, and Equal variance (LINE)

- L: a linear relationship exists between Y and the X's.
- I: the data are independent.
- N: the residuals are normally distributed.
- E: the variance of the residuals is equal for all X's.

Assessing Model Fit: Residual Analysis

- Residuals = observed − predicted values

$$e_i = Y_i - \hat{Y}_i$$

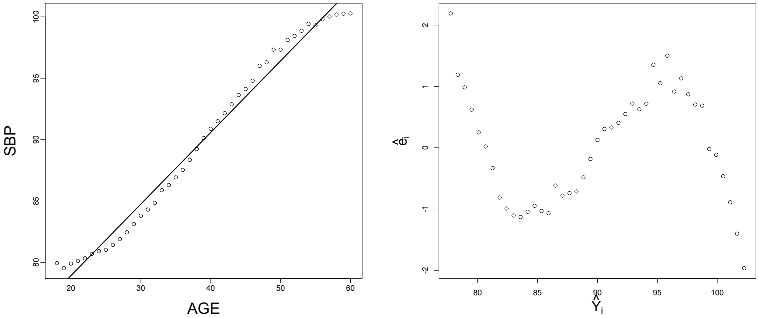
- Plots to examine:
  - Residuals vs. fitted: check linearity & homoscedasticity
  - Q-Q plot: check normality
  - Residuals vs. time/order: check independence
- Look for:
  - No clear pattern in plots
  - Residuals centered around 0

Line

- Assumption: there is a linear relationship between Y and the X's. I.e.,
  - The expected value of the residual is zero for all combinations of X's
  - $E(e_i) = 0$.
- How to check
  - Plot Y_i vs. X_i.
  - Plot e_i vs. $\hat{Y}_i$

Line

Example Violation:



Corrective actions:

- Fit a different regression model.
- Add quadratic of cubic components.
- Transformation of Y and/or X

Normality

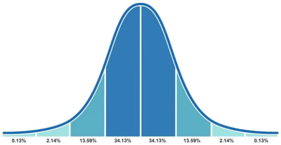
- Assumption: The residuals are normally distributed.
- How to check
  - Histogram of the e_i's.
  - Normal probability plot of e_i.
  - Tests of normality on e_i.
  - Outlier tests.

Normal Distribution

- The **Normal (Gaussian) distribution** for a continuous random variable X:

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Parameters:
  - μ: mean (center)
  - σ^2: variance (spread)
- Symmetric, bell-shaped curve.
- Appears naturally in many public health applications (e.g., measurement error, heights, blood pressure).



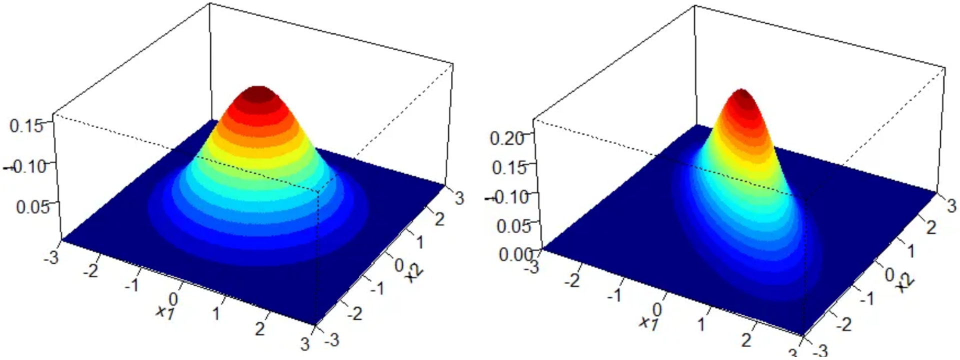
Multivariate Normal Distribution

- A random vector $\mathbf{X} \in \mathbb{R}^p$ follows a **Multivariate Normal distribution** if

$$f(\mathbf{x} \mid \boldsymbol{\mu}, \Sigma) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

- Parameters:
  - $\boldsymbol{\mu} \in \mathbb{R}^p$: mean vector
  - Σ: $p \times p$ covariance matrix
- Contours are ellipses in 2D, ellipsoids in higher dimensions.
- Correlations between variables are captured through Σ.

Multivariate Normal Distribution

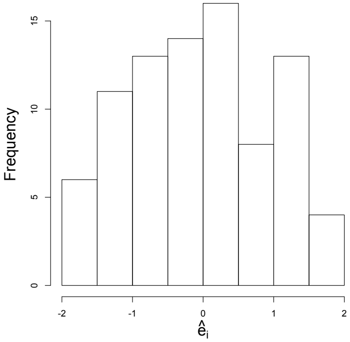


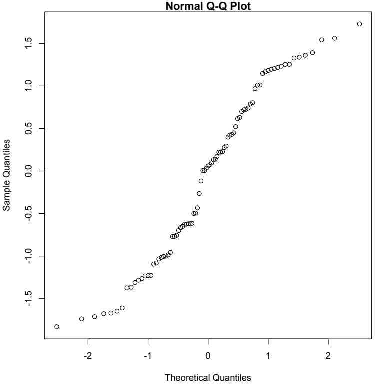
Normality

Example Violation:

Corrective actions:

- Examine outliers to determine if they are contaminated
- Transformation of Y and/or X

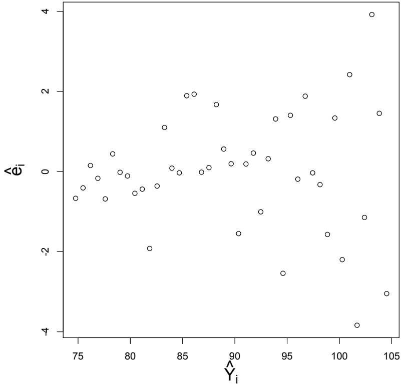




Equal Variance

- **Assumption:** the variance of the residuals is equal for all X's.
- How to check
  - Plot Y_i vs. X_i.
  - Plot e_i vs. $\hat{Y}_i$

  Example Violation $\Longrightarrow$



- **Corrective action:**
  - Transformation of Y.
  - Patterned covariance model

Visual Checks for Assumptions

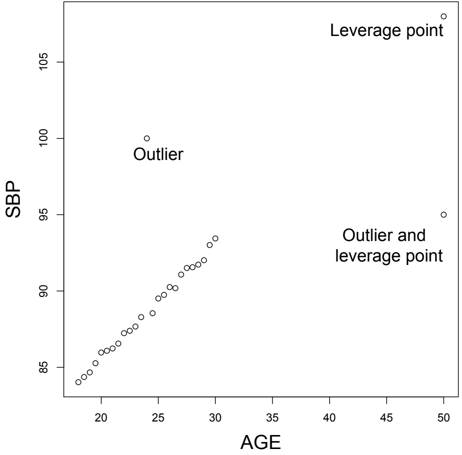
- **Linearity:** Scatterplots of Y vs. X, or residuals vs. fitted values
- **Independence:** Time series plots (look for patterns)
- **Normality:** Histogram or Q-Q plot of residuals
- **Equal Variance:** Residuals vs. fitted plot (look for “funnel” shape)

Residual analysis is your main tool to diagnose model issues.

Identifying influential points

We will discuss the detection of **outliers** and **leverage points**.

- What's the difference?
  - **Leverage point:** unusual predictor combination, i.e. $(X_{i1}, \ldots, X_{ip})$.
  - **Outlier:** Unusual Y_i values.
- h_{ii} the ith diagonal element of $H = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$ measures leverage.



Methods to identify influential points

- Cook's Distance (Cook's D) – measures the influence of a point on the estimated regression line
- Cook's D utilizes h_{ii} and e_i. It quantifies:
  - How much the prediction would change if observation i is removed
  - The larger the Cook's D, the more influential the point to the estimated regression line
  - Empirical cutoff point for Cook's D: $4/n$
- Notes
  - a point with a high leverage does not have to be an influential point (can have small residuals)
  - a point with a high leverage and a large residual is likely to be an influential point

Properties of $\hat{\boldsymbol{y}}$, β and σ^2

- The variance of $\hat{\boldsymbol{\beta}}$ is

$$Var(\hat{\boldsymbol{\beta}}) = (\boldsymbol{X}'\boldsymbol{X})^{-1}\sigma^2$$

  where σ^2 is estimated using

$$\hat{\sigma}^2 = \frac{1}{n-p-1}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

- Neither of these results requires any additional assumptions.

Properties of $\hat{\boldsymbol{y}}$, β and σ^2

Once we impose the LINE assumptions, we get the following properties

- $\hat{\boldsymbol{\beta}} \sim MVN(\boldsymbol{\beta}, (X'X)^{-1}\sigma^2)$, hypothesis tests for $\hat{\beta}_j$ can be based on

$$T_j = \frac{\hat{\beta}_j}{\sqrt{\hat{\sigma}^2 v_j}} \sim t_{n-p-1}$$

  where v_j is the jth diagonal element of $(X'X)^{-1}$.
- $(1-\alpha) \times 100\%$ confidence intervals can be made via

$$\hat{\beta}_j \pm t_{n-p-1}(\alpha/2)\sqrt{\hat{\sigma}^2 v_j}$$

Why Model Selection?

1. *Prediction accuracy:* to make reasonable predictions or estimations, we need
   - **accuracy** $\rightarrow$ on average, what we estimate is equal to what we expect (e.g., $\hat{Y}$ in the long run is equal to the population mean of Y)
   - **precision** $\rightarrow$ small variation in prediction/estimation
2. *Interpretation:* if we can limit the number of variables, we can better understand the "main factors" driving the outcome.

For this section, we will assume that the assumptions of the model are met and focus on finding the best model.

Prediction Case Study: Hospital Length of Stay (LOS)

- **Goal (prediction):** Predict continuous **LOS (days)** at admission using routinely available EHR features.
- **Why it matters:** bed management, staffing, discharge planning, and cost forecasting.
- **Typical predictors:** demographics, admission source, vitals/labs at ED, comorbidities/CCI, service line, prior utilization, time-of-day/weekday.
- **Outcome:** LOS in days (often right-skewed; consider log-transform or robust loss).

Almeida, G., Brito Correia, F., Borges, A. R., & Bernardino, J. (2024). Hospital length-of-stay prediction using machine learning algorithms—a literature review. *Applied Sciences*, 14(22), 10523.

Baseline Model and Diagnostics

Baseline: Multiple linear regression

$$\text{LOS}_i = \beta_0 + \boldsymbol{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$$

- Start with domain-driven covariates (demographics, vitals, comorbidities).
- Check skew / heavy tails: consider $\log(\text{LOS} + 1)$; inspect residuals.
- **Leakage audit:** exclude post-admission or post-discharge info (e.g., final procedures, discharge destination decided later), and engineer predictors only from data *available at prediction time*.

Metrics: RMSE/MAE on hold-out; calibration plot of predicted vs. observed LOS; error by service line.

Best-Subset Selection

- Best-subset can look over all 2^p models or look at it for only those with k variables.
- For the latter, this method finds for each k the subset of k variables that minimizes the RSS.
- Even for p as large as 30 or 40 there are algorithms to estimate this quickly.
- This procedure cannot be used to estimate k (RSS will necessarily go down with k)
- To choose k we need to use some other criteria.

Model Selection Criteria

1. MSE, i.e., the estimate of σ^2.
   - If the model underfits the data $\rightarrow$ large residuals $\rightarrow$ large SSE $\rightarrow$ large MSE.
   - If the model overfits the data $\rightarrow$ df for the error part is lost $\rightarrow$ large MSE
   - A good model is expected to have a relatively small MSE.
2. Adjusted R^2 (R^2_{Adj} in SAS, or $\bar{R}^2$)
   - $\bar{R}^2 = 1 - \frac{MSE}{SS_{TOTAL}/(n-1)}$
   - This is similar to R^2, but we are "punished" (via df) for using too many terms in the model
   - Since SS_{TOTAL} is not changed between models, as MSE decreases, $\bar{R}^2$ increases.
   - In SAS, use "selection=adjrsq" in PROC REG if want to use this criterion for model selection

$*$ The two criteria MSE and $\bar{R}^2$ are equivalent.

Model Selection Criteria

3. Mallow's C(p) or C_p
   - This criterion looks at both accuracy and precision.
   - p: number of terms in the model under consideration, $p \leq k$ (k: the number of independent variables in the full model).
   - $\hat{\sigma}^2$: MSE from the full model (with k independent variables).
   - $MSE(p)$: MSE from the model under consideration.
   - Def. of $C(p)$

$$C(p) = SSE(p)/\hat{\sigma}^2 - [n - 2(p+1)]$$

   - a good model has C(p) close to (p+1)
   - Mallow's Cp is equivalent to Akaike information criterion (AIC).

Model Selection Criteria

4. PRESS (prediction SS)
    - For linear regression, calculating the predicted values when removing one subject is relatively simple (you don't have to re-estimate β).
    - The PRESS statistic is a leave-one-out (LOO)-CV.
    - A good model has a small PRESS statistic.
5. Cross-validation of the data $\rightarrow$ the concept is similar to PRESS. Not really needed since PRESS is easy to calculate.

Automated Model Selection Procedures

- Useful when there are a large number of possible independent variables.
- Can use these procedures to get a manageable number of "candidate models".
- These procedures start with a full or null model and automatically add or subtract variables.
- Computationally, these methods are cheap.
- We don't look at as many models (more constrained search): results will have lower variance, but possibly more bias.
- They depend on criteria to evaluate models, traditional software will use p-values (which is not recommended) here we'll consider AIC.

Automated Model Selection Procedures

1. Forward selection
    a. begin with nothing (except for the intercept) in the model
    b. enter the variable with the highest R^2 (SLR) (assuming it increases AIC over a null model)
    c. determine which variable would result in the highest increase in AIC and add it to the active set of variables
    e. repeat step c until no more variables increase the AIC
2. Backward selection
    a. begin with all variables in the model
    b. Delete the variable that would result in the highest increase in AIC
    c. repeat step b until no variable would improve AIC

Automated Model Selection Procedures

3. Stepwise selection
   a. Enter the variable with the highest R^2 (SLR) (assuming it increases AIC over a null model)
   b. At each step, check which variable would result in the highest increase in AIC and add it to the active set of variables
   c. check if removing any variable would improve the AIC, if there is such a variable remove it from the active set
   d. repeat steps b and c until no variable improves AIC or a loop is encountered.

Forward-Stagewise Regression

- The Forward-Stagewise Regression algorithm can be considered as a "slow" version of the forward selection algorithm.[1]
- This algorithm is closely related to the LASSO method we'll discuss later.
- Therefore, a thorough understanding of this method is essential as we proceed.
- For this algorithm (and many others), we'll assume that the $\boldsymbol{X}$ vector has been standardized.

[1] See Efron, Hastie, Johnstone, and Tibshirani (2003) "Least Angle Regression" (with discussion) *Annals of Statistics* for a good description of this method.

The Forward-Stagewise Regression algorithm

1. Let $\hat{\boldsymbol{y}}^{(0)} = \bar{y}$, $\hat{\boldsymbol{\beta}}^{(0)} = \mathbf{0}$, and $\boldsymbol{r}^{(0)} = \boldsymbol{y} - \hat{\boldsymbol{y}}^{(0)}$
2. For step $k = 1, 2, \ldots$
   2.1 let $\boldsymbol{c}^{(k)} = \boldsymbol{X}'(\boldsymbol{y} - \hat{\boldsymbol{y}}^{(k-1)})$ be the vector of *current correlations*
   2.2 find $j^* = \text{argmax}_j |c_j^{(k)}|$ be the $\boldsymbol{X}$ variable with the largest current correlation.
   2.3 Let $\delta = \epsilon \cdot \text{sign}(c_{j^*}^{(k)})$and set

$$
\begin{aligned}
\hat{\beta}_{j^*}^{(k)} &= \hat{\beta}_{j^*}^{(k-1)} + \delta \\
\hat{\boldsymbol{y}}^{(k)} &= \hat{\boldsymbol{y}}^{(k-1)} + \delta \boldsymbol{X}_{j^*}
\end{aligned}
$$

   where ϵ is a "small" constant.

   repeat step 2 until $\max_j |c_j^{(k)}| \leq \kappa$

The Forward-Stagewise Regression algorithm notes

- If $\kappa = 0$ the algorithm will continue until we reach the LS estimates
- The choice of ϵ (the step-size) is important.
- This procedure can take many more than p steps.
- The basic idea is to combine many "weak" models (i.e., ϵ is small) to build a powerful "committee" model
- That is, we find the variable that best describes the data to the residuals from our current model.
- Later, we'll discuss **boosting** (a common machine learning approach), which has similarities with this algorithm.